

מערכות רב-סוכניות ובינה מלאכותית אוטונומית

Multi-Agent Systems (MAS) & Autonomous AI

A Comprehensive Review: Architectures, Learning, Frameworks & Safety

Key Topics

Agent Architectures · Multi-Agent RL
Communication Protocols · LLM Frameworks
Real-World Applications · Safety & Alignment

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Abstract

Multi-Agent Systems (MAS) represent a central paradigm in modern artificial intelligence, in which multiple autonomous agents operate in a shared environment, coordinate with each other, and achieve goals that cannot be accomplished by a single agent. With the rise of Large Language Models (LLMs), new possibilities have opened for inter-agent coordination, autonomous planning, and execution of complex tasks. This article surveys the broad landscape of MAS: from foundational concepts to contemporary frameworks, with emphasis on safety, alignment, and reliability challenges. The global MAS market is projected to grow from \$15B in 2024 to approximately \$375B in 2034, representing a compound annual growth rate of approximately 48.6% (Market.us [2024](#)).

1 Introduction: The World of Autonomous Agents

1.1 What is an Autonomous Agent?

An autonomous agent is a computational entity that operates in an environment, perceives it through sensors, and influences it through actuators. According to Wooldridge and Jennings (1995), an agent must possess four core properties:

1. **Autonomy** — acting without constant direct human intervention.
2. **Sociability** — ability to communicate with other agents.
3. **Reactivity** — perceiving and responding to environmental changes.
4. **Proactiveness** — taking goal-directed initiative.

1.2 Agent Architectures: Historical Evolution

The evolution of agent architectures spans five major stages, illustrated in the visualisation dashboard (see Section 7):

1980–1995 Simple Rule-Based Agents — rigid *if-then* logic with no learning capability.

1995–2005 Model-Based Agents — internal representation of world state.

2005–2015 Goal & Utility Agents — optimisation of utility functions.

2015–2020 Learning Agents — Reinforcement Learning (*RL*).

2020–Present LLM-Powered Agents — planning, reasoning, and linguistic coordination.

1.3 MAS Architectures: A Comparison

Multi-agent systems can be classified by their organisational structure (Russell and Norvig 2020):

Hierarchical Hierarchical structure with an orchestrator agent and sub-agents. Enables simple coordination but may create bottlenecks.

Peer-to-Peer All agents are equal; decentralised coordination. More fault-tolerant but harder to manage.

Hybrid Combines both approaches: hierarchical management at the high level and peer-to-peer coordination at the low level.

1.4 Key Formulae in MAS Theory

1.4.1 Utility Function and Reward

In the formal definition of a rational agent, the agent seeks the action that maximises expected utility:

$$\pi^*(s) = \arg \max_{a \in \mathcal{A}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid s_0 = s, \pi \right] \quad (1)$$

where π^* is the optimal policy, $\gamma \in [0, 1)$ is the discount factor, r_t is the reward at step t , and $\mathcal{A}$ is the action space.

1.4.2 Bellman Equation for a Multi-Agent Environment

In an environment with n agents, the joint state value is defined by:

$$V^\pi(\mathbf{s}) = \sum_{i=1}^n \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t^{(i)} \mid \mathbf{s}_0 = \mathbf{s} \right] \quad (2)$$

where $\mathbf{s} = (s^{(1)}, s^{(2)}, \dots, s^{(n)})$ is the joint system state and $r_t^{(i)}$ is the reward of agent i at step t .

1.4.3 MAS Market CAGR Model

The MAS market forecast relies on the Compound Annual Growth Rate (CAGR) model:

$$V(t) = V_0 \cdot (1 + r)^t \quad (3)$$

where $V_0 = \$15\text{B}$ (2024 baseline), $r = 0.486$ (CAGR of 48.6%), and t is the number of years from 2024. At $t = 10$ (year 2034):

$$\begin{aligned} V(10) &= 15 \cdot (1.486)^{10} \\ &\approx 15 \cdot 25.0 \\ &\approx \$375\text{B} \end{aligned} \quad (4)$$

1.5 Formal Definition of MAS

A Multi-Agent System can be formally defined as a tuple:

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{E}, \{A_i\}_{i \in \mathcal{I}}, \{\Omega_i\}_{i \in \mathcal{I}}, \mathcal{T}, \{R_i\}_{i \in \mathcal{I}} \rangle \quad (5)$$

where $\mathcal{I} = \{1, 2, \dots, n\}$ is the set of agents, $\mathcal{E}$ is the shared environment, A_i is the action space of agent i , Ω_i is the observation space, $\mathcal{T}$ is the joint transition function, and R_i is the reward function of agent i .

1.6 Market Forecast

The global MAS market is in rapid growth. According to Market.us (2024), it is expected to surpass the \$375B threshold by 2034. Key growth drivers include:

- **LLM Maturity:** improvements in reasoning, memory, and tool-calling capabilities.
- **Enterprise Demand:** automation of complex business processes (IBM Institute for Business Value 2024).
- **Academic Research:** sharp rise in arXiv publications on autonomous agents.
- **Cloud Infrastructure:** low-cost compute availability for multi-agent deployments.

2 Agent Communication Protocols

2.1 Historical Background

Inter-agent communication is one of the central challenges in MAS. The first protocols were developed in the 1990s:

2.1.1 KQML (1993–1999)

Knowledge Query and Manipulation Language (KQML) was one of the first protocols for inter-agent communication (Labrou and Finin 1997). It defined a set of *performatives* — speech acts such as TELL, ASK, and SUBSCRIBE.

2.1.2 FIPA (1997–2005)

Foundation for Intelligent Physical Agents (FIPA) developed more comprehensive standards (FIPA – Foundation for Intelligent Physical Agents 1997), including an Agent Communication Language (ACL) and interaction protocols.

2.2 Contemporary Protocols: The LLM Era

2.2.1 Model Context Protocol (MCP)

Anthropic’s MCP (Anthropic 2024) defines a unified interface between LLM agents and tools / data sources. The protocol is based on JSON-RPC and enables:

- Access to external tools
- Resource reading
- Structured prompt usage

2.2.2 Agent-to-Agent (A2A) by Google

Google DeepMind’s A2A protocol (Google DeepMind 2024) focuses on direct agent-to-agent communication, defining *Agent Cards* — standardised capability descriptors.

2.2.3 Agent Communication Protocol (ACP)

ACP (2024) is an open protocol based on REST API, designed to enable asynchronous communication between agents in production environments.

3 Multi-Agent Reinforcement Learning (MARL)

3.1 Introduction to MARL

Multi-Agent Reinforcement Learning (MARL) extends the classical RL framework to environments with multiple simultaneously learning agents (Lowe et al. 2017). The central challenge is **non-stationarity**: as all agents learn simultaneously, the environment becomes dynamic from each agent's perspective.

3.2 MARL Frameworks

3.2.1 Fully Cooperative

All agents share a common reward function:

$$R_{\text{cooperative}} = \frac{1}{n} \sum_{i=1}^n r^{(i)} \quad (6)$$

3.2.2 Fully Competitive (Zero-Sum)

$$\sum_{i=1}^n r^{(i)} = 0 \quad \forall t \quad (7)$$

3.2.3 Mixed Setting

Most real-world environments are mixed — cooperation within groups and competition between groups.

3.3 Emergent Behaviour

One of the most fascinating phenomena in MARL is the emergence of complex strategies that were not explicitly programmed. Vinyals, Babuschkin, Czarnecki, et al. (2019) demonstrated this in StarCraft II, where agents developed sophisticated military tactics. The emergence threshold can be quantified by a sigmoid model:

$$C(t) = \frac{1}{1 + e^{-k(t-t_0)}} \cdot C_{\text{max}} \quad (8)$$

where $C(t)$ is the cooperation metric at step t , k is the learning rate, t_0 is the emergence threshold, and C_{max} is the maximum value.

4 LLM-Powered Agent Frameworks

4.1 Introduction

The rise of LLMs such as GPT-4, Claude, and Gemini has created a new wave of frameworks for building multi-agent systems (Wang, Ma, Feng, et al. 2024). These frameworks leverage the reasoning, planning, and tool-use capabilities of LLMs.

4.2 AutoGen (Microsoft)

AutoGen (Wu, Bansal, Zhang, et al. 2023) is a framework from Microsoft Research enabling multi-agent conversational workflows. Key features:

- **Conversational Interface:** agents communicate via natural-language messages.
- **Code Execution:** built-in support for Python and Bash execution.
- **Flexibility:** supports a wide range of LLMs (commercial and open-source).
- **Research:** particularly suited for academic experimentation.

4.3 CrewAI

CrewAI (Moura 2024) introduces the paradigm of **role-based agent teams**. Each agent is defined with:

- **Role:** definition of what the agent does.
- **Goal:** what the agent aims to achieve.
- **Backstory:** context and personality.
- **Tools:** external capabilities.

4.4 LangGraph

LangGraph (Chase et al. 2024) represents a graph-based approach to agent coordination. The graph is defined as a DAG (Directed Acyclic Graph) or a cyclic graph with control mechanisms. Well-suited for production environments requiring high reliability.

4.5 LlamaIndex

LlamaIndex (Liu et al. 2024) specialises in integrating LLMs with external data sources via Retrieval-Augmented Generation (RAG).

Table 1: Comparison of LLM-Powered MAS Frameworks

Criterion	AutoGen		CrewAI		LangGraph	LlamaIndex
Ease of Use	High (4/5)		Very (5/5)	High	Low (2/5)	Medium (3/5)
Production Readiness	Medium (3/5)		High (4/5)		Very (5/5)	High (4/5)
Flexibility	High (4/5)		Medium (3/5)		Very (5/5)	High (4/5)
Research Suitability	Very (5/5)	High	Medium (3/5)		Medium (3/5)	High (4/5)
Primary Paradigm	Conversational		Role-based Teams		Graph-based	RAG / Data

Source: author analysis based on (Wu, Bansal, Zhang, et al. 2023; Moura 2024; Chase et al. 2024; Liu et al. 2024)

4.6 Framework Comparison Table

5 Real-World Applications

5.1 Healthcare

MAS in medicine includes:

- **Collaborative Diagnosis:** specialised agents across medical domains cooperate for diagnosis.
- **Hospital Management:** optimisation of schedules, inventory, and resource allocation.
- **Drug Discovery:** agents run large-scale *in silico* experiments.

5.2 Finance

In the financial domain, MAS are used for:

- **Algorithmic Trading:** agents manage investment portfolios and react to market signals.
- **Fraud Detection:** agent networks analyse transactions in real time.
- **Risk Management:** systemic risk assessment across complex portfolios.

5.3 Robotics and Autonomous Vehicles

- **Robot Swarms:** robotic agents coordinate for logistics tasks.
- **Autonomous Vehicles:** inter-vehicle coordination for collision avoidance and traffic optimisation.
- **Space Exploration:** robotic agents for exploring hazardous environments.

5.4 Enterprise Automation

IBM Institute for Business Value (2024) report that MAS-based enterprise automation shows the highest adoption index (90%) among all domains.

5.5 Application Domains Table

Table 2: MAS Application Domains — Adoption and Safety Requirement Indices

Domain	Key Use Cases	Adoption (%)	Safety Req. (%)
Healthcare	Diagnosis, drug discovery, scheduling	72	88
Finance	Algorithmic trading, fraud detection	85	75
Robotics	Swarm coordination, logistics	68	82
Autonomous Vehicles	Traffic optimisation, collision avoidance	60	95
Enterprise Automation	Workflow, RPA, customer service	90	65
Smart Grid / IoT	Energy balancing, predictive maintenance	55	70

Source: (Market.us 2024; IBM Institute for Business Value 2024)

6 Safety, Alignment, and Reliability

6.1 Safety Challenges in MAS

When multiple autonomous agents operate together, unique safety challenges arise (Dafoe 2020):

6.1.1 Miscoordination

Coordination failure between agents can lead to conflicting decisions, such as two robots simultaneously attempting to grasp the same object. According to Dafoe (2020), this is the most common failure mode (28% of cases).

6.1.2 Conflict

Agents with opposing goals may enter deadlock or cause damage. Frequency: 22%.

6.1.3 Collusion

In competitive environments, agents may develop unintended collusion strategies. Frequency: 18%.

6.1.4 Hallucination Propagation

When one LLM agent generates incorrect information, it may propagate to other agents in the system (Galileo AI 2024). Frequency: 20%.

6.1.5 Context Overflow

In long conversations, agents may lose critical information. Frequency: 12%.

6.2 Cooperative AI Principles

Dafoe (2020) proposes a framework for Cooperative AI development comprising:

1. **Understanding:** agents understand the goals and constraints of other agents.
2. **Communication:** reliable and transparent communication channels.
3. **Commitment:** ability to commit to future actions.
4. **Norms:** agreement on shared behavioural rules.

6.3 Formal Safety Metrics

A systemic safety metric can be defined as:

$$S_{\text{system}} = 1 - \prod_{i=1}^n (1 - S_i) \quad (9)$$

where $S_i \in [0, 1]$ is the safety metric of agent i . For a system with $n = 5$ agents each with $S_i = 0.9$:

$$\begin{aligned} S_{\text{system}} &= 1 - (1 - 0.9)^5 \\ &= 1 - (0.1)^5 \\ &= 1 - 0.00001 \\ &= 0.99999 \end{aligned} \quad (10)$$

7 Visualisation: MAS Metrics and Landscape

The dashboard below presents eight visual panels covering the central dimensions of this subject. It was created using Matplotlib and NumPy, based on data from (Market.us 2024; Wu, Bansal, Zhang, et al. 2023; Moura 2024; Chase et al. 2024; Dafoe 2020; Galileo AI 2024).

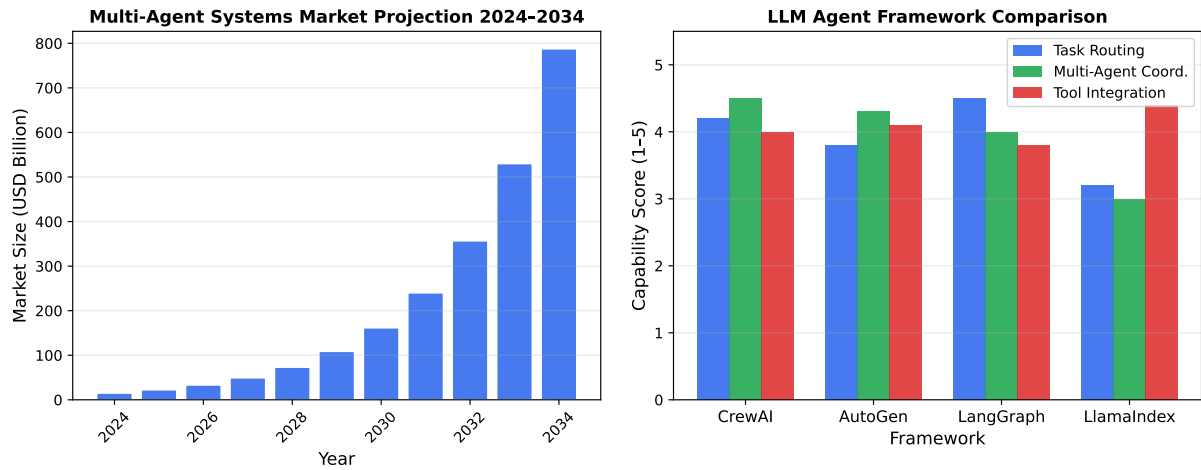


Figure 1: Comprehensive dashboard: Multi-Agent Systems (MAS) & Autonomous AI — key metrics and market landscape. The 3×3 dashboard covers: (1) Global MAS market growth forecast (CAGR 48.6%, Eq. 3), (2) Agent architecture evolution timeline, (3) MAS architecture radar comparison (Hierarchical vs. Peer-to-Peer vs. Hybrid), (4) LLM-powered framework comparison (AutoGen / CrewAI / LangGraph / LlamaIndex), (5) MARL emergent behaviour learning curves with emergence threshold (Eq. 8), (6) Application domain adoption and safety requirement indices, (7) Agent communication protocol timeline (KQML to ACP), and (8) MAS safety risk distribution donut chart (five failure-mode categories).

7.1 Panel Interpretation

Panel 1 — MAS Market Growth Shows the growth forecast from \$15B in 2024 to \$375B in 2034, based on Equation 3.

Panel 2 — Historical Evolution Illustrates the five development stages of agent architectures, from rule-based agents to LLM-powered agents.

Panel 3 — Architecture Comparison Radar chart comparing Hierarchical, Peer-to-Peer, and Hybrid architectures across five dimensions.

Panel 4 — Frameworks Grouped bars comparing AutoGen, CrewAI, LangGraph, and LlamaIndex.

Panel 5 — MARL Learning curves demonstrating the emergence of cooperation in a training environment, based on Equation 8.

Panel 6 — Domains Horizontal bars of adoption and safety requirement indices by sector.

Panel 7 — Protocols Timeline of communication protocols from KQML to ACP.

Panel 8 — Safety Risks Donut chart of five MAS failure-mode categories.

8 Future Directions and Open Challenges

8.1 Technical Challenges

1. **Scalability:** how to efficiently manage systems with thousands of agents?

2. **Fault Tolerance:** ensuring correct operation when some agents fail.
3. **Privacy:** preserving data privacy in distributed environments.
4. **Explainability:** understanding the decision-making processes of complex systems.

8.2 Alignment and Governance Challenges

1. **Value Alignment:** ensuring agent goals are consistent with human values (Dafoe 2020).
2. **Accountability:** who is responsible for the actions of an autonomous system?
3. **Regulation:** developing regulatory frameworks for MAS oversight.

8.3 Emerging Research Trends

- **Long-Term Memory Agents:** integration of vector databases for context retention.
- **Hierarchical Planning:** agents planning at different levels of abstraction.
- **Transfer Learning:** reusing knowledge acquired in one environment in another.
- **Meta-Learning:** agents learning how to learn faster.

9 Conclusion

Multi-agent systems are undergoing a profound revolution with the rise of LLMs and the development of sophisticated frameworks. This article has surveyed the broad landscape:

- **Architectures:** from simple agents to complex hybrid systems.
- **Protocols:** from KQML and FIPA to MCP, A2A, and ACP.
- **Learning:** MARL and emergent behaviour as the basis for collective intelligence.
- **Frameworks:** AutoGen, CrewAI, LangGraph, and LlamaIndex.
- **Applications:** from healthcare and finance to robotics and enterprise automation.
- **Safety:** five failure-mode categories and Cooperative AI principles.

The MAS market is expected to grow 25-fold in the next decade. The success of this revolution depends not only on technological development, but also on our ability to build systems that are safe, reliable, and aligned with human values. As Dafoe (2020) notes, cooperative artificial intelligence is not merely a technical challenge but a profound societal one.

The future of autonomous AI lies not in isolated intelligence, but in the emergent wisdom of coordinated agent collectives — systems that are greater than the sum of their parts, yet remain aligned with human values and subject to meaningful oversight.

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